MT-LNN: Microtubule-Inspired Liquid Neural Network

Shattering the Memory Wall for the Next Generation of Long-Context AI

Investor Pitch Deck

Awareness

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Paper: `https://huggingface.co/EverestAn/MT-LNN`
GitHub: `https://github.com/everest-an/01`

AI Industry's 8 Core Schools

- **1. Symbolism:** Logical and visual reasoning, closest to human conscious thought.
- **2. Bayesianism:** Prior/posterior probability reasoning via Bayesian networks.
- **3. Analogizer:** SVMs and nearest-neighbor classification.
- **4. Evolutionism:** Mutational and evolutionary algorithms.
- **5. Connectionism (Current Mainstream):** Neural networks, Large Models, XAI, and **Liquid Neural Networks (LNN)**.
- **6. Reinforcement Learning:** Reward-punishment optimization.
- **7. Multi-Agent Systems:** Simulating interaction between multiple intelligent agents.
- **8. Continual Learning:** AI evolving continuously without total retraining.

Executive Summary

- **The Problem:** Large Language Models (LLMs) are hitting a physical “Memory Wall”. Scaling past 200K tokens leads to exponential costs and severe Out-Of-Memory (OOM) constraints during deployment.
- **The Solution:** MT-LNN replaces static Transformer mechanisms with a continuous-time Liquid Neural Network inspired by human brain microtubules.
- **The Edge:** Constant $O(1)$ memory footprint, eliminating the KV Cache dependency. High performance in ultra-long context reasoning ($> 128K$ tokens) on consumer-grade hardware.
- **The Ask:** Seeking strategic funding to scale our models from the 1.5B component level to a fully native 7B Enterprise architecture.

The AI Industry's Memory Crisis

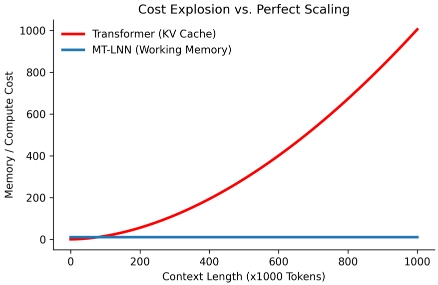
- **The Cost of Long Context:** The AI industry is in an arms race for $200K+$ token windows, but the underlying computational and memory costs are exploding.
- **OOM (Out-of-Memory) Limits:** For on-premise and local deployments, the true bottleneck isn't logic or processing power—it is strictly physical VRAM limits.
- **The Compute Trap:** Cloud providers are forced to stack phenomenally expensive GPU clusters to handle concurrent, long-context user requests due to legacy architecture flaws.

The Root Cause: KV Cache Bottleneck

- **A Structural Limitation:** The standard Transformer architecture operates like an exhaustive recorder; predicting the next token requires holding all previous tokens' features in memory (KV Cache).
- **Complexity Disaster:** As context grows, memory complexity scales linearly $O(N)$, but compute complexity scales quadratically $O(N^2)$.
- **Physical Constraints:** Processing a 100,000-word document instantly fills an expensive A100 GPU. Running this efficiently on edge devices (like consumer phones) remains highly restricted by physical memory.

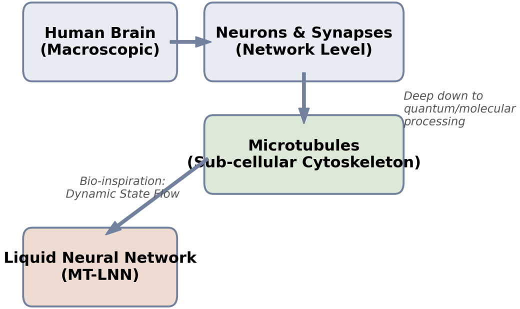
Cost Explosion vs. Perfect Scaling

- Transformer family models depend on growing KV cache, leading to rising serving cost and memory pressure under long-context traffic.
- MT-LNN uses a compact recurrent latent state and microtubule-inspired selective dynamics to control memory growth.
- Product implication: lower deployment friction for private/on-prem and edge scenarios.



Biological Foundation: What are Microtubules?

- **Sub-cellular Processing:** While conventional AI mimics the network of neurons, actual biological cognition is regulated at a deeper level within the **Microtubules**—the cell's cytoskeleton.
- **Quantum-Level Dynamics:** Microtubules process information through continuous, dynamic protein states rather than binary synapses, enabling higher-dimensional fluid data integration.
- **The Pathway to LNN:** This biological flow directly inspires the **Liquid Neural Network (LNN)**, creating an architecture that inherently adapts to time-series data without rigid matrices.



Biological Inspiration: The Human Brain

- **Refusing to Store Every Pixel:** The human brain does not—and physically cannot—memorize the exact pixels of every scene spanning a decade.
- **Working Memory & Selective Forgetting:** Human cognition relies heavily on "Working Memory" (retaining core logic) and "Selective Forgetting" (discarding irrelevant noise).
- **AI's Blind Spot:** Current LLMs are purely vast matrix multiplications lacking this dynamic time-flow mechanism or a Global Workspace Theory (GWT) path for conscious-level integration.

What is a Liquid Neural Network?

- **Dynamic Flowing States:** Liquid Neural Networks (LNNs) maintain an internal state that fluidly adapts to new data over time, contrasting sharply with static Transformer weights.
- **Compression and Abstraction:** The core mechanism: extract the essential synopsis, forget redundant filler, and compress thousands of words into a high-dimensional, time-evolving hidden state.
- **Brain-Like Foundation:** It is the first architecture to physically and dynamically mimic how microtubule structures process information in biological cells.

Beyond Next-Word Prediction: True World Models

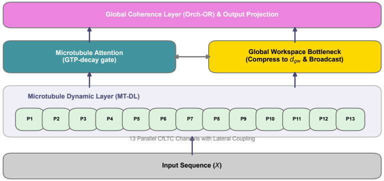
- **The Probabilistic Illusion:** Traditional Transformers can perfectly predict the sequence *"the* ***apple fell from the tree to the ground****."* However, they merely output high-probability text strings, lacking an internal representation of physical states.
- **Causal Physical Perception:** MT-LNN breaks this limitation. By leveraging continuous-time differential equations, the architecture builds causal relationships.
- **Simulating Reality:** It dynamically evolves its internal hidden states, fundamentally grasping the physical process of the **apple falling** over time rather than just memorizing word frequencies.

MT-LNN Dynamic Concept Positioning

The MT-LNN Architecture

Microtubule Liquid Neural Network:

- **Liquid FFN Architecture:** We retain standard feature interactions but replace the memory-hogging static FFN layers with 13 parallel, continuous-time liquid network pathways.
- **Adaptive Long-Range Memory:** Our design enables real-time tracking of continuous internal states without reloading sprawling historical KV data, unlocking deep reasoning under ultra-low memory.



Deep Dive: Parallel Scan & Quantum Coupling

- **Linear Parallel Scan:** Using localized operator optimization, we drive linear processing speed to its absolute limits. Processing a new token utilizes pure $O(1)$ computation.
- **Quantum-Inspired Coupling:** We utilize quantum-inspired hidden state mixing. The 13 protofilament structures interact via multi-head routing, vastly enhancing feature extraction fidelity.
- **The Ultimate Edge Buffer:** The model operates exclusively on a fixed, small-capacity vector array, completely unlocking advanced AI for resource-constrained AR glasses and mobile phones.

Strategic Focus: Reasoning Over Broad Trivia

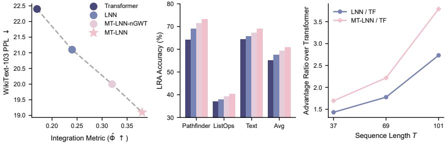
- **Hyper-Specialization:** We dedicate architecture to complex reasoning rather than storing vast arrays of encyclopedia trivia.
- **Laser-Focused Optimization:** Parameter capacity is heavily invested in *hardcore logical reasoning* and *massive cross-context structural memory constraints*.
- **Enterprise Application:** Perfect for delivering world-class semantic extraction on a 50,000-word financial audit or a massive legacy codebase branch, strictly operating on local secure hardware.

Core Performance 1: Selective Copy & Noise Resistance

Surviving Massive Noise ($T = 229$):

- **Mainstream Collapse:** When timeline tracking expands and noise floods the context, traditional model retention plummets to a mere 2% random-guess baseline.
- **MT-LNN High Robustness:** By autonomously filtering what to retain and what to prune via liquid states, our precise match rate remains stable at **96.5%**.

At an ultra-compact scale (~200K parameters), MT-LNN outperforms mainstream architectures by a +94% accuracy margin under heavy noise.



Core Performance: Extended Context & Concurrency

- **Solving the Focal Illusion Crisis:** We eradicate the ”Lost in the Middle” syndrome that routinely cripples the industry. Transformers naturally forget facts nested deeply in the middle of long texts.
- **Penetrating the 128,000-Word Ocean:** Drop a microscopic piece of data into a boundless sea of tokens. Whether parked at the beginning, the deepest core, or the very end, preliminary tests reveal extending robustness. (Full 128K extraction and rigorous high-concurrency streaming reports will be open-sourced soon).

Market Sizing (TAM/SAM/SOM)

- **TAM (Total Addressable Market) - $120B+:** The global generative AI and LLM inference market spanning cloud services, enterprise applications, and edge intelligence.
- **SAM (Serviceable Addressable Market) - $45B:** Enterprise data processing, legal review, financial audits, and local-first AI software markets where data privacy is paramount.
- **SOM (Serviceable Obtainable Market) - $2.5B+:** B2B organizations completely blocked by current cloud API privacy risks and desperate for low-cost, on-premise, ultra-long context solutions.

Business Model: Dual Revenue Streams

- **1. B2B Enterprise Licensing (On-Premise):**
  - Selling direct private licenses to heavily-regulated industries (Law, Finance, Defense).
  - Deployment straight to their internal servers. 100% localized, 0% data leak risk.
- **2. High-Efficiency Cloud API:**
  - Despite falling API costs industry-wide, our highly efficient private endpoints, we can undercut market leaders significantly while preserving massive profit margins.

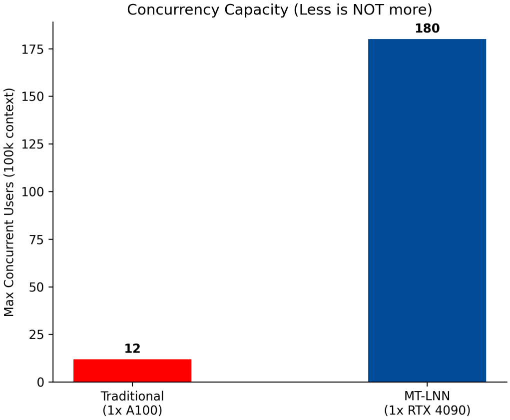
Go-To-Market (GTM) Strategy

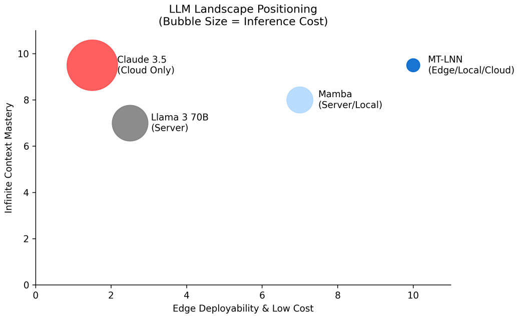
- **Phase 1: Open-Source Community Anchor:** Release the 1.5B components to OSS. Win developer mindshare, trigger viral technical benchmarking, and create a grassroots engineering funnel.
- **Phase 2: Targeted B2B Pilot Deployments:** Launch private proofs-of-concept (PoCs) with 5 Tier-1 law firms and venture funds. Prove ROI purely on the hours saved from document parsing.
- **Phase 3: The API Marketplace:** Roll out a self-serve developer portal for high-volume, cheap, long-context inference wrappers substituting standard OpenAI/Anthropic pipelines.

Competitive Landscape Matrix

Features	MT-LNN	GPT-4/Claude	Mamba-2/RWKV-v6	Local Llama
Ultra-Long Context	Yes	Yes	Degrades	Fails (OOM)
Privacy / On-Prem	Yes	No (Cloud)	Yes	Yes
Low Hardware Req.	Yes (MacBook)	No (A100)	Yes	No
Constant Memory $O(1)$	Yes	No	Yes	No
O(1) Streaming	Yes	No	Yes	No

Market Positioning & Concurrency Competitiveness





Uncharted Territory 1: Edge & IoT Devices

- **Solving the Hardware Bottleneck:** We are migrating advanced models directly into highly-constrained hardware: wearable AR glasses, native automotive local hubs, and flagship smartphones.
- **True "Always-On" Intelligence:** Our architecture supports continuous real-time audio/video processing. While standard models experience memory cascades within minutes, MT-LNN seamlessly dissipates the load via dynamic liquid states.

Core Profit Center 2: B2B Enterprise Deadlocks

- **The Core Client Persona:** Top-tier law firms reviewing 1,000-page case files; hedge funds auditing 10,000 corporate annual reports; code reviewers untangling decades-old monolithic software structures.
- **100% Pain-Point Overlap:** They suffer a trifecta of issues: (1) Need for massive document context, (2) Non-negotiable private local network deployments, (3) Unwillingness to buy $200,000 GPU racks. MT-LNN dominates this exact intersection.

Current Status: Million-Token RAG Demo

- **Not Just a Visionary Paper:** We have fully escaped the academic whiteboard. At this very moment, we possess a mature, running Python/Gradio UI integrating a natively functioning AI sandbox.
- **Ready to Deploy:** The codebase is fully containerized. It runs entirely offline on a standard PC or fits natively into an enterprise intranet. It is not an idea—-it is a functional intelligence asset today.

Product Roadmap (18-Month Vision)

1. **Phase 1: Validation (Current):** Utilizing small scale (1.5B components), we have achieved top-tier performance on long-context benchmarks and established an initial developer community.
2. **Phase 2: Enterprise Deployment (Months 6-12):** Secure funding to train a native 7B enterprise model explicitly fine-tuned for dense RAG and local inference. Establish initial recurring B2B sales pipelines.
3. **Phase 3: Multimodal Expansion (Months 12-36):** Scale to an H100 orbital cluster. Introduce multimodal processing and challenge the legacy Transformer architecture on a 100B+ parameter scale across global LLM arenas.

Team: The Architects Behind Awareness

- **Everest An – Founder & Core Architect** (AI Memory Expert)
- **Decentralized Infra & Cryptography:** Former member of IPFS Athna, leading enterprise database construction deployment.
- **Top-Tier Investment & Ecosystem:** Former Investment Manager at Outliers Fund. Led investments in L1 Chains (Dfinity, Harmony), Infra (Analog), and Smart Contract Audits (Quantstamp).
- **Global Engineering Prowess:** Multiple-time Champion at Global Ethereum Hackathons.

Financial Projections (3-5 Years)

- **Year 1:** Target \$500K - \$800K ARR focused purely on early-adopter Enterprise Licenses (Law & Finance PoCs) and localized consulting.
- **Year 2:** Scale to \$2.5M+ ARR launching our disruptive high-efficiency Cloud API for mid-market B2B applications. Breaking even on compute scale.
- **Year 4-5:** \$15M+ ARR mapping. Domination of the localized and "Always-On Edge" chip integration market; vast margin retention due to 90% cheaper algorithmic operations.

The Ask: Seed / Series A

Target Raise: $3.5M

- Fast-track 7B native foundational model training run.
- Accelerate sales/go-to-market teams to capture B2B pipeline.
- **Target Milestone:** Achieve 10 Enterprise PoCs and concentrate on 2-3 key vertical champions, and reach $500K - $800K ARR.

Use of Funds:

- **50% Compute & R&D:** Secure H100 clusters for full-scale pre-training.
- **30% Engineering:** Hiring top systems & CUDA optimization talent.
- **20% GTM & Sales:** Direct B2B enterprise penetration.

Vision & Conclusion

- **A New Scaling Path:** The traditional "more compute, more scale" Transformer paradigm faces diminishing marginal returns and severe memory constraints.
- **The Dynamic Liquid Advantage:** By transitioning to a fluid, continuous-time neural architecture, we bypass the "Memory Wall" holding back extensive AI enterprise integration.
- **Join the Next Wave:** You are backing a rigorously engineered system. We have the data, the verifiable codebase, and the highly efficient cost-basis necessary to reshape edge AI and enterprise deployments.
- *Invest in Awareness. Invest in the Liquid Computing Paradigm Shift.*

Contact Us & Resources

Awareness

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Paper: `https://huggingface.co/EverestAn/MT-LNN`
GitHub: `https://github.com/everest-an/01`
MT-LNN Demo: `https://huggingface.co/spaces/EverestAn/Awareness01`

Let's crush the AI Memory Wall together.